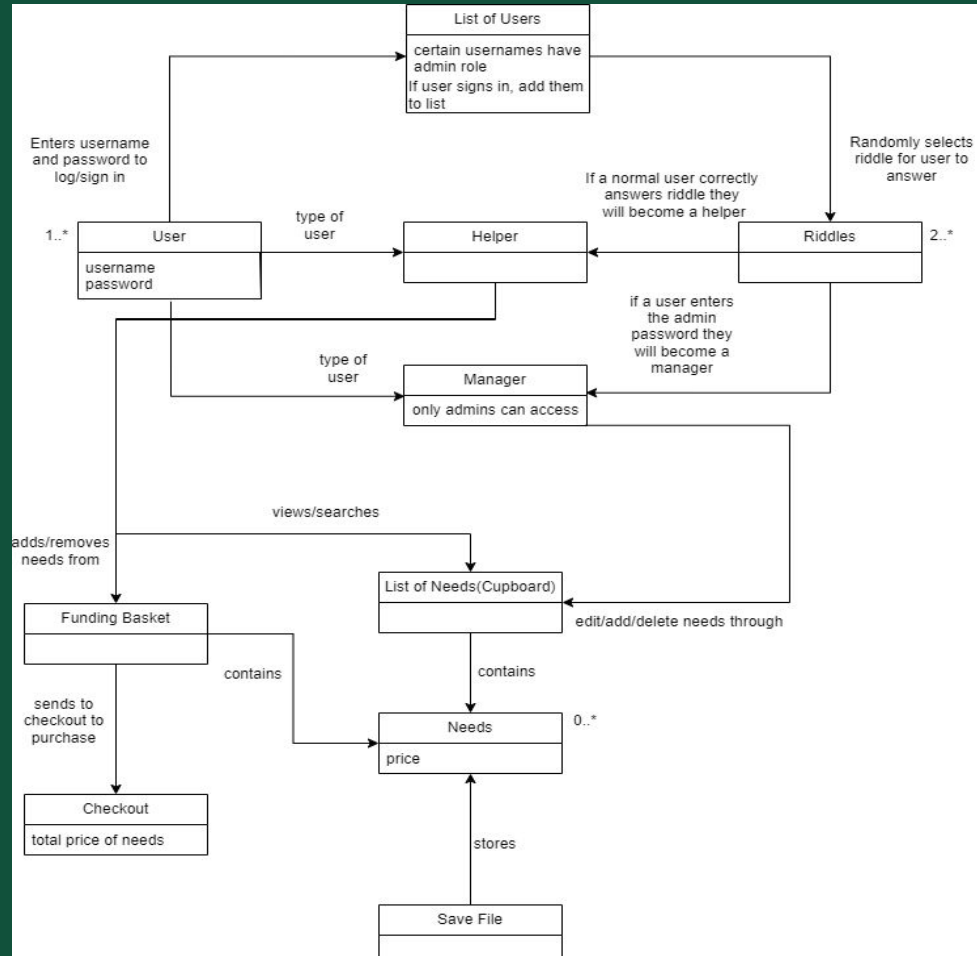


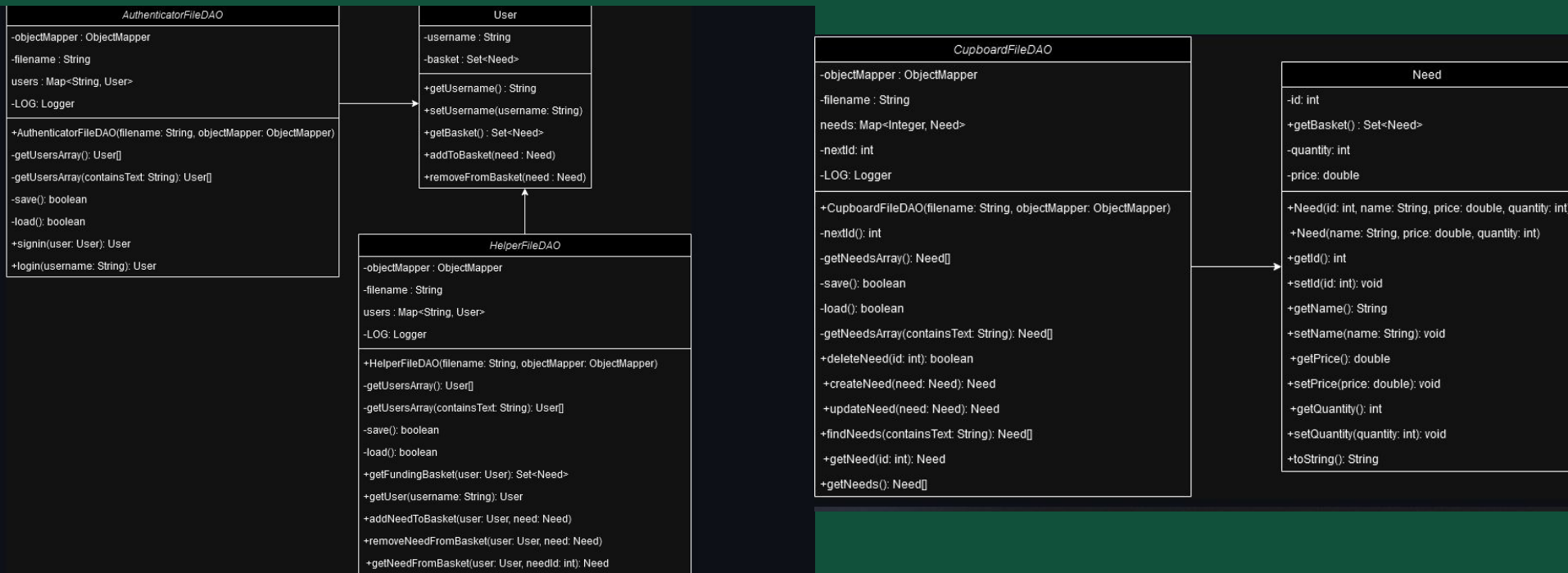
MONEY FOR MASTERMINDS

By: Yat Long Chan, Ben Hemmers, Guancheng Liu,
Graden Olson, Vincent Son

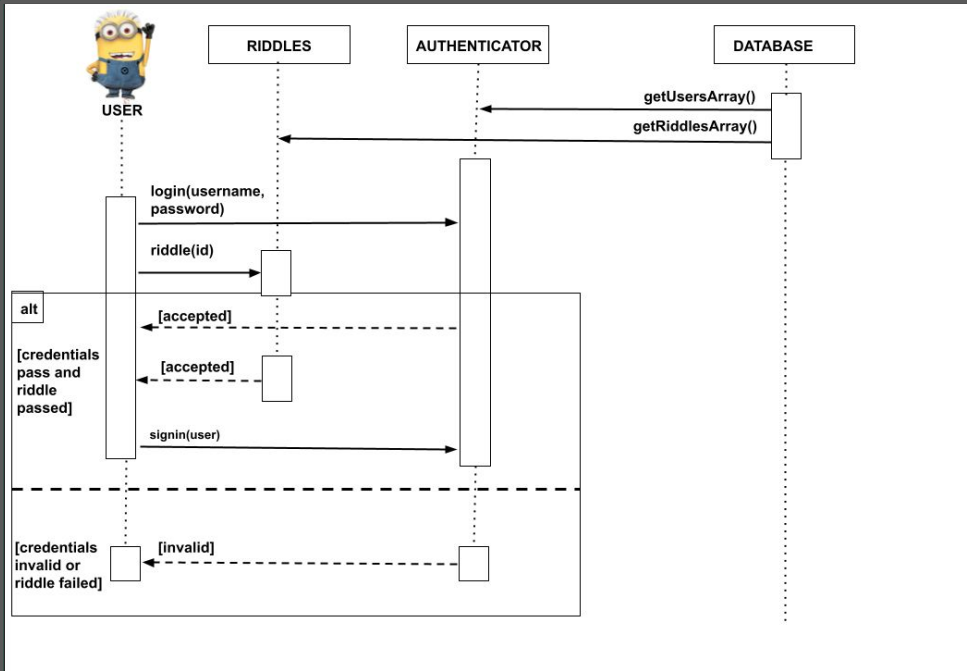
Final domain model



Architectural overview of the tiers and any subsystems within tiers



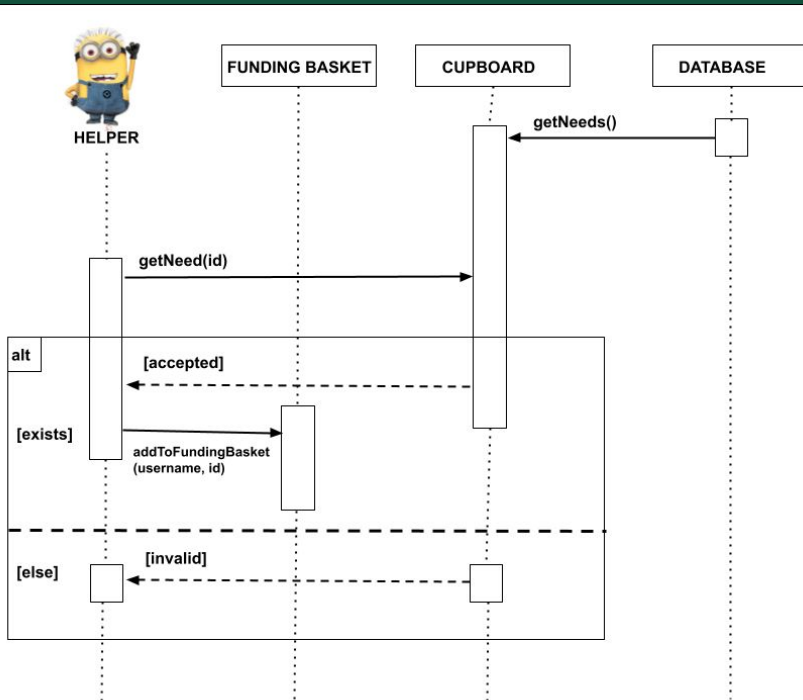
Major Class and Sequence Diagrams



Authentication Sequence Diagram

- Authenticator retrieves a random riddle and valid users
- User sends `login(username, password)` → Authenticator, `riddle(id)` → Riddles
- If credentials && riddles pass → accepted...
 - `signin(user)`
- Otherwise...
 - return invalid

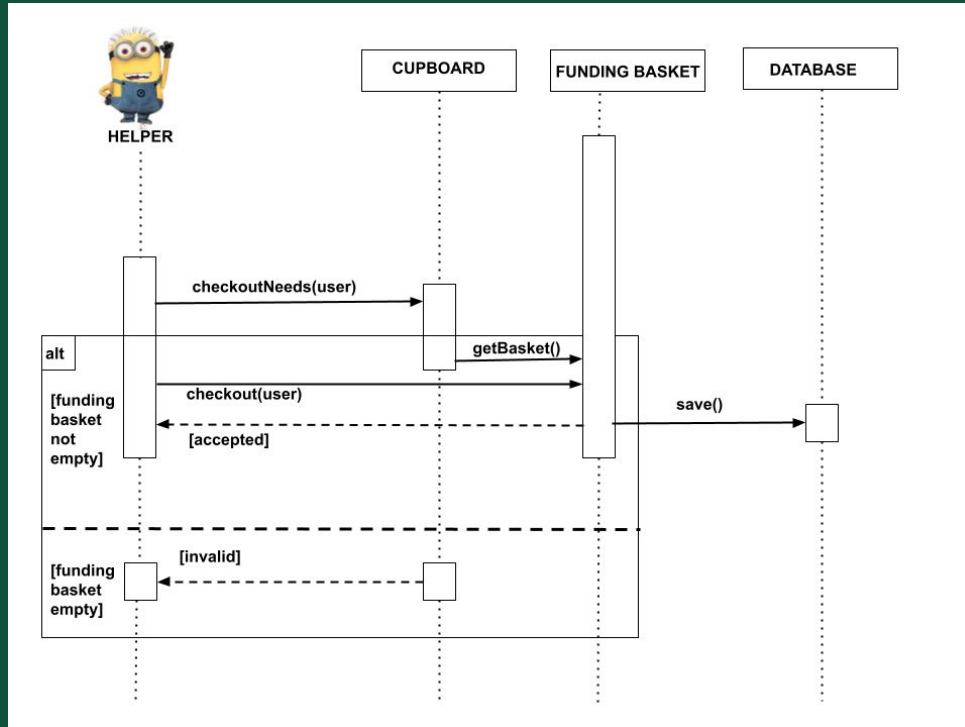
Major Class and Sequence Diagrams



Helper Sequence Diagram

- Cupboard needs retrieved from database with `getNeeds()`
- User sends `getNeed(id) → Cupboard`
- If exists $\rightarrow$ accepted...
 - `addToFundingBasket(username, id)`
- Otherwise...
 - `return invalid`

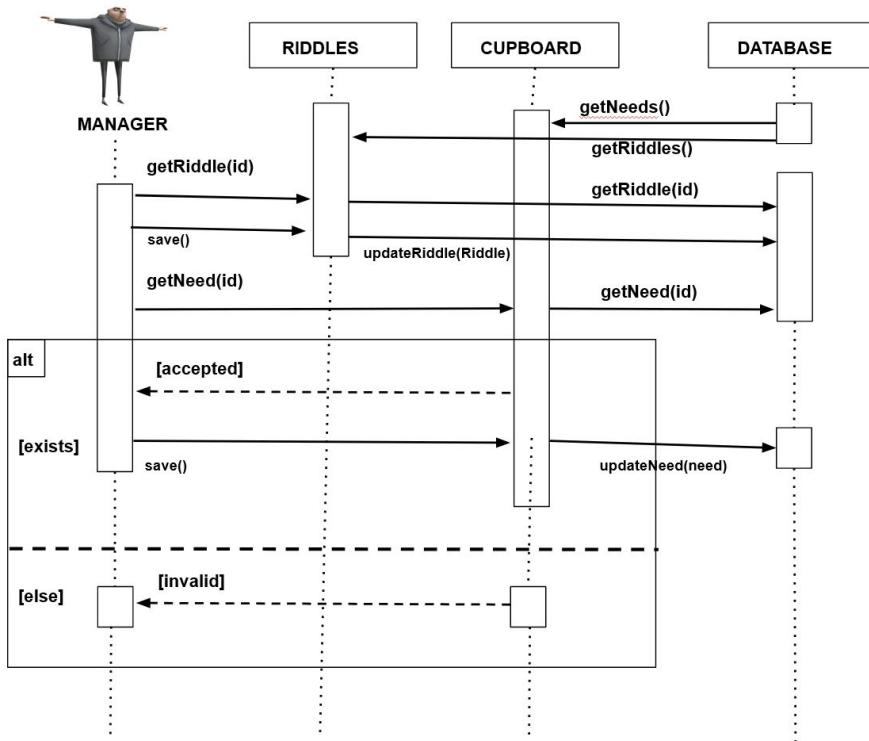
Major Class and Sequence Diagrams



Checkout Sequence Diagram

- `checkoutNeeds(user)` prompted to Cupboard first, then Funding Basket
- If the funding basket is already empty, fails
- Otherwise...
 - Successfully checks out the helper's funding basket
 - Updates database

Major Class and Sequence Diagrams

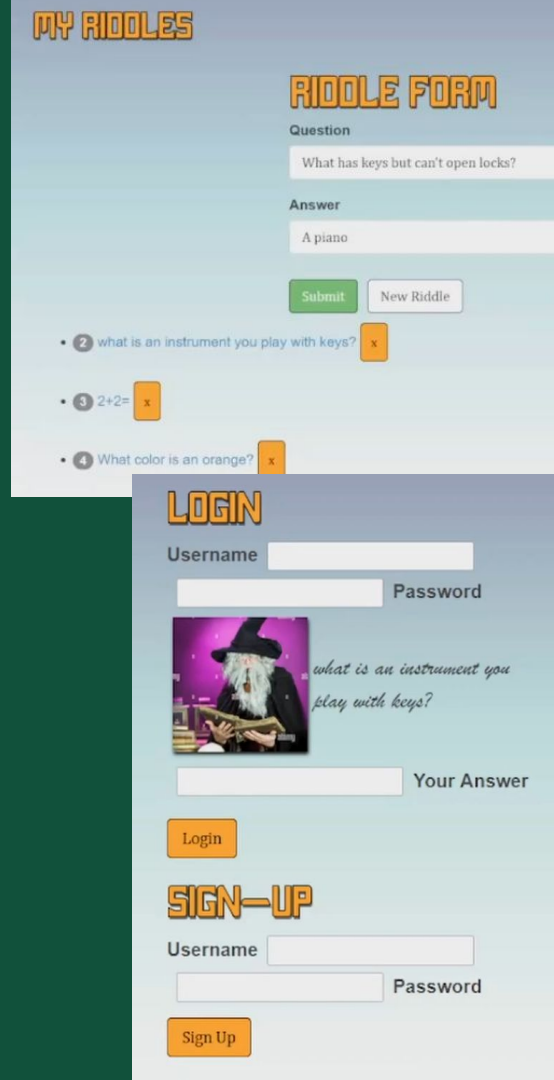


Manager Sequence Diagram

- Manager can create/delete/edit riddles, retrieved from database with `getRiddles()`
- Manager can update needs retrieved from database with `getNeeds()`
- If need exists...
 - `updateNeed(need) → Database`
- Otherwise...
 - `return invalid`

Highlight strong areas of our project

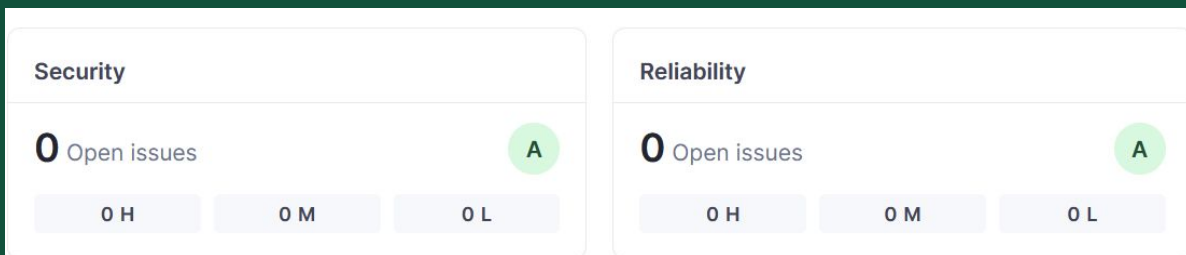
- Authentication (riddles, password)
- Handling user access to pages



Static Code Analysis highlights

Java Code:

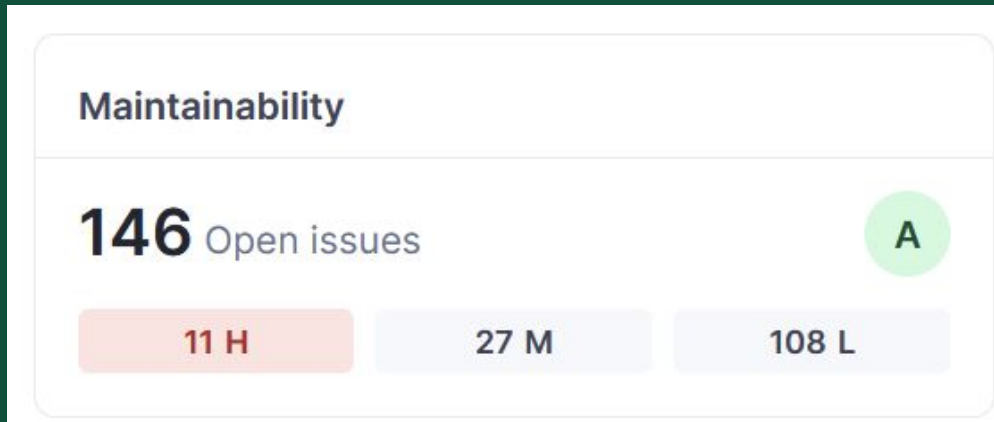
- Overall reliable and secure



Static Code Analysis highlights

Java Code:

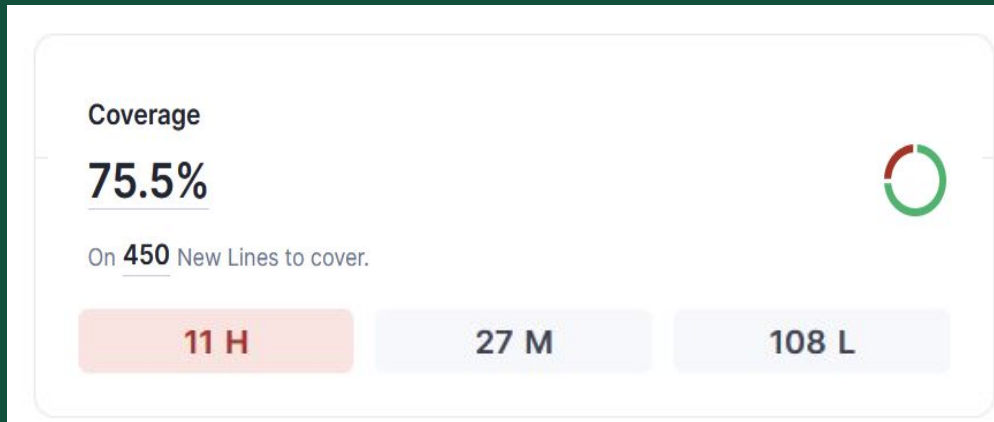
- Overall reliable and secure
- Maintainability issues



Static Code Analysis highlights

Java Code:

- Overall reliable and secure
- Maintainability issues
- Improve code coverage



Static Code Analysis highlights

Java Code:

- Overall reliable and secure
- Maintainability issues
- Improve code coverage
- Security Risk?

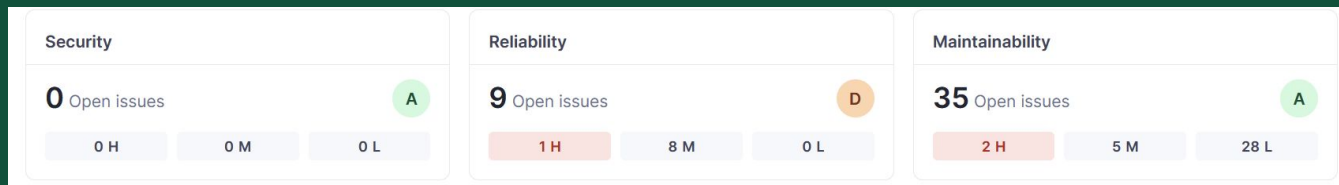
```
}  
int randomIndex = (int)(Math.random() * riddles.size());
```

Make sure that using this pseudorandom number generator is safe here.

Static Code Analysis highlights

Angular Code:

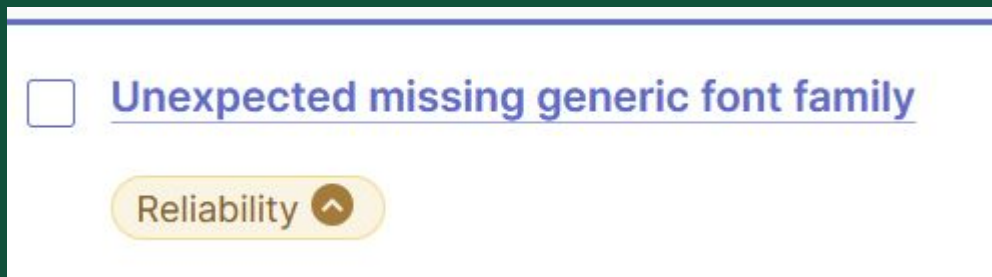
- Reliability issues



Static Code Analysis highlights

Angular Code:

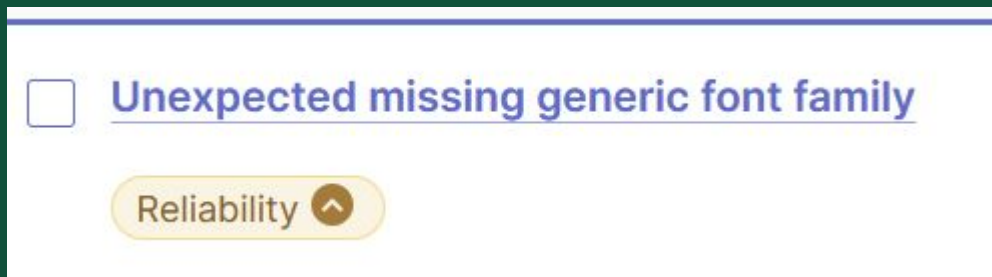
- Reliability issues
 - Missing generic font family



Static Code Analysis highlights

Angular Code:

- Reliability issues
 - Missing generic font family



Static Code Analysis highlights

Angular Code:

- Reliability issues
 - Missing generic font family
 - Fill out / remove function

```
getCheckedoutNeeds(): void {
```

Refactor this getter so that it actually refers to the property 'checkedoutNeeds'.

Static Code Analysis highlights

Angular Code:

- Reliability issues
 - Missing generic font family
 - Fill out function
- Maintainability issues
 - Unused imports
 - Unfilled functions

Remove this unused import of 'Observable'.

Maintainability ▼

☐ Unexpected empty method 'getCheckedoutNeeds'.

Maintainability

Intentionality

suspicious +

Open

Not assigned

L23 • 5min effort • 17 days ago • Code Smell • Critical

☐ Unexpected empty method 'checkout'.

Maintainability

Intentionality

suspicious +

Open

Not assigned

L26 • 5min effort • 17 days ago • Code Smell • Critical

Recommendations for software architecture design improvements

- Reduce number of parts in API (we have a lot of services and controllers)
- Design components to be loosely coupled, with minimal dependencies between them, allowing for easier changes and maintenance.
- Make changes to CSS sheets to reduce redundancy

Address any future usability design improvements

- Move sign up to a different page
- Save sessions across multiple instances of the web application
- Additional security
 - Password hashing in case heroes hack into the server
 - Implements email for multi-factor authentication and for forgetting password.
- Improve UI
 - Better use of white space, bigger buttons



LOGIN

Username

Password



what is an instrument you
play with keys?

Your Answer

Login

SIGN-UP

Username

Password

Sign Up

LOGIN

Username

Password



what is an instrument you
play with keys?

Your Answer

Login

SIGN-UP

Username

Password

Sign Up

MY NEEDS

NEED FORM

Name

Price

Quantity

MY NEEDS

MONEY FOR MASTERMINDS

DASHBOARD CUPBOARD RIDDLES LOGOUT

MY RIDDLES

RIDDLE FORM

Question
What has keys but can't open locks?

Answer
A piano

Submit New Riddle

- 1 what is an instrument you play with keys?
- 2 2+2=
- 3 What color is an orange?

MONEY FOR MASTERMINDS

DASHBOARD CUPBOARD

NEED FORM

Name

Price

Quantity

Submit New Need

MONEY FOR MASTERMINDS

DASHBOARD CUPBOARD RIDDLES LOGOUT

TOP NEEDS

SUPER SUIT

LASER

SPACE SHIP

Need Search

is
super suit
Laser

LOGIN

Username Bubby

Password



what is an instrument you
play with keys?



Solve
Spider-M

MY FUNDING BASKET

- 15 LASER
- 13 SUPER SUIT
- 12 RAY GUN